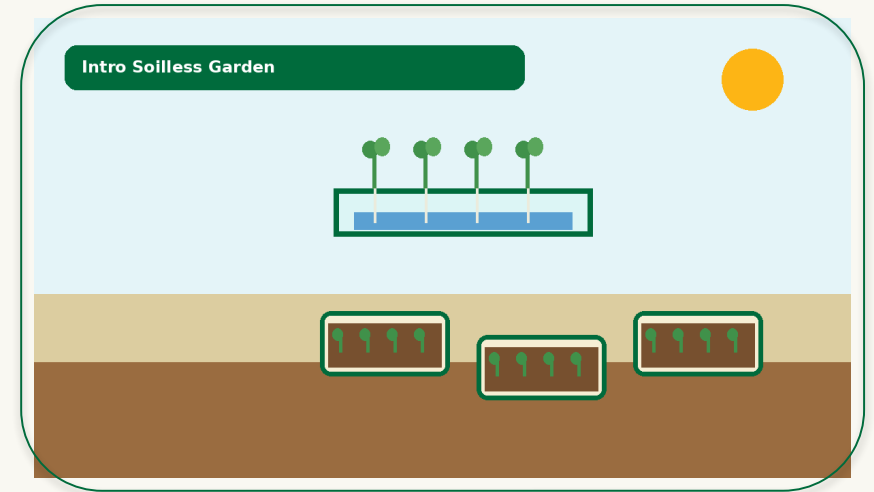


Intro Soiless Garden

Growing Plants Without Soil

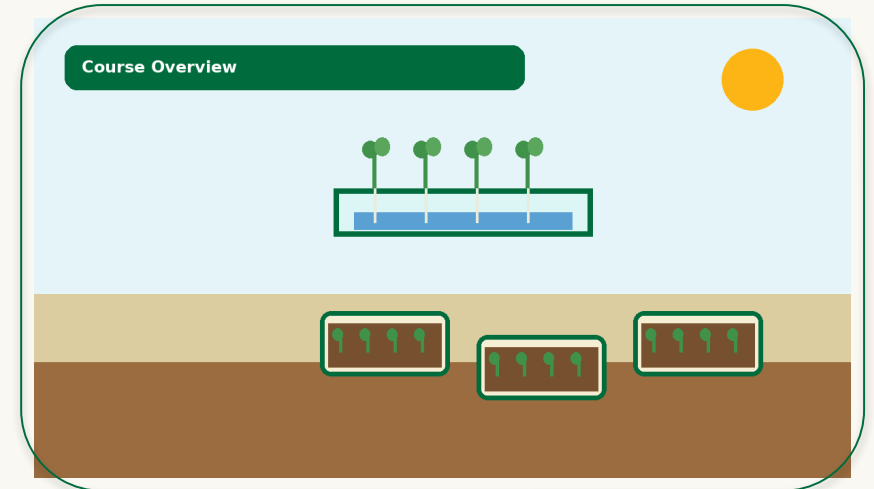
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Learning progression: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system and its educational purpose; teach all ten technical topics at Primary 4-6 level; guide safe practical work and measurement; connect science, mathematics and agriculture; maintain useful records; help pupils interpret observations without overclaiming.
- Core materials: teacher guide, system diagram, child-safe tools, labels, clipboards, rulers/tapes, graduated containers or measuring devices where relevant, observation/data sheets, clean harvest containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson, measurement or investigation plan.



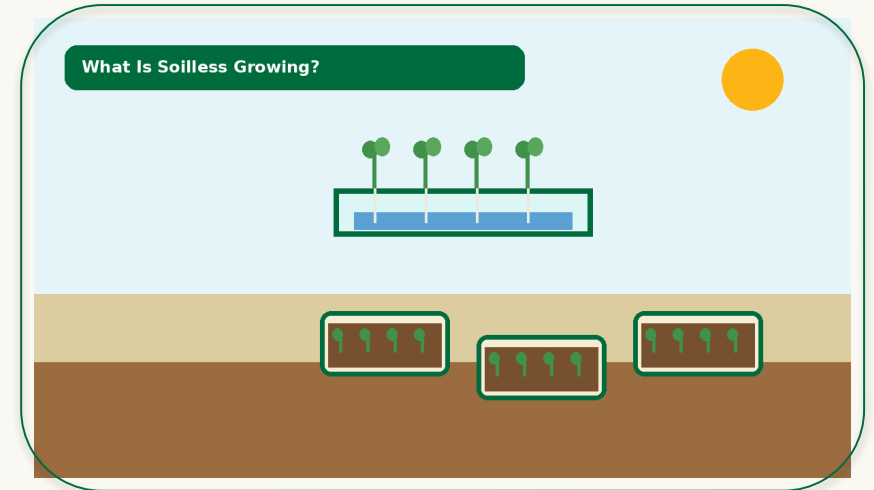
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

What Is Soilless Growing?

Teacher knowledge and practical application

- Define soilless growing as producing plants without field soil, using water and/or other root-support media.
- Explain that roots still need water, oxygen, nutrients and suitable conditions.
- Show the installed school system before introducing terminology.
- Distinguish soilless growing from simply placing a plant in water.
- Keep systems school-scale and supervised.



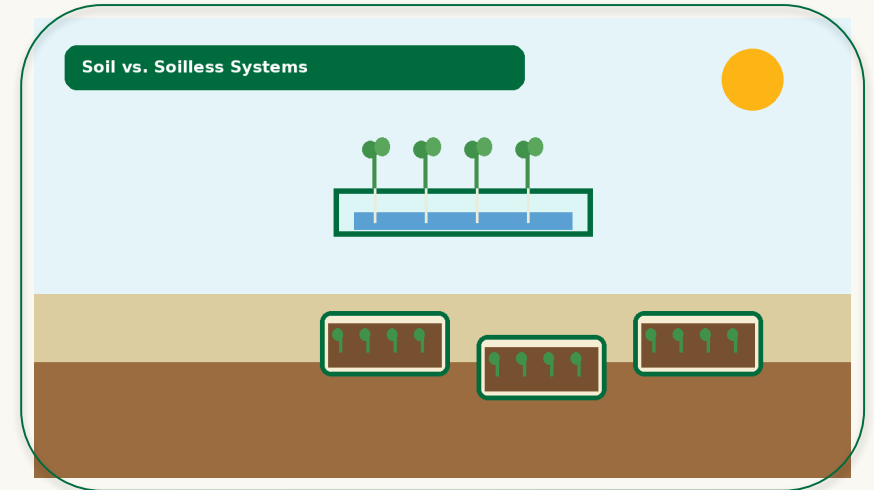
TEACHER PRACTICE

- Prepare a labeled soilless-system orientation.

Soil vs. Soilless Systems

Teacher knowledge and practical application

- Compare where roots are supported, how water is supplied and how nutrients reach plants.
- Avoid presenting either system as universally better.
- Use the same crop where possible for a fair classroom comparison.
- Discuss differences in maintenance and monitoring.
- Record observations rather than relying on assumptions.



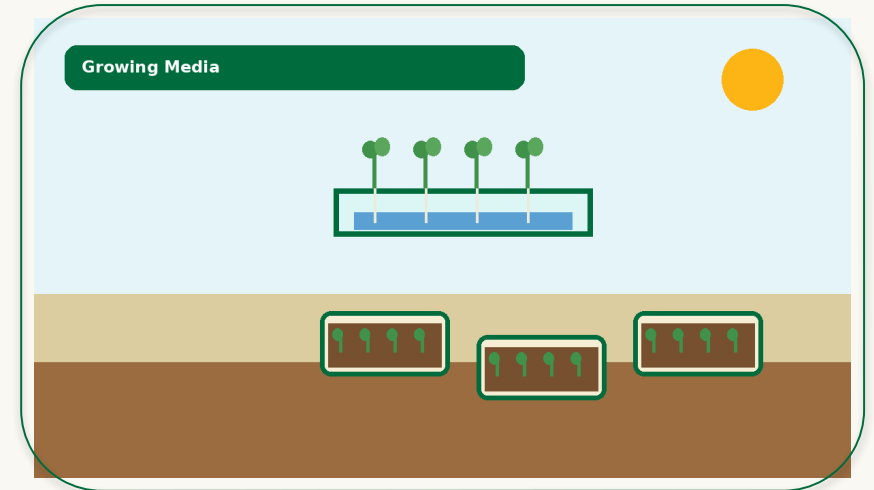
TEACHER PRACTICE

- Create a soil-versus-soilless comparison table.

Growing Media

Teacher knowledge and practical application

- Introduce approved media such as coco coir, perlite or other materials used by the installed system.
- Explain that media can support roots and hold water/air but may contribute little nutrition.
- Use clean materials only.
- Avoid inhalation of dusty media and follow handling guidance.
- Do not substitute untested waste materials.



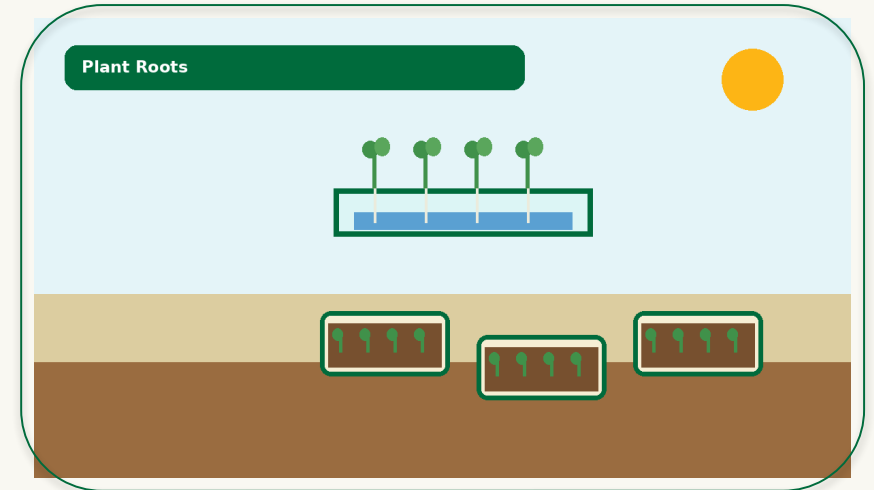
TEACHER PRACTICE

- Prepare safe samples of the media used in the school system.

Plant Roots

Teacher knowledge and practical application

- Observe healthy roots through accessible system sections or planned samples.
- Discuss roots as structures for anchorage and uptake of water and nutrients.
- Keep roots protected from unnecessary light and damage where the system requires it.
- Do not repeatedly remove plants for inspection.
- Compare root environments in soil and soilless systems.



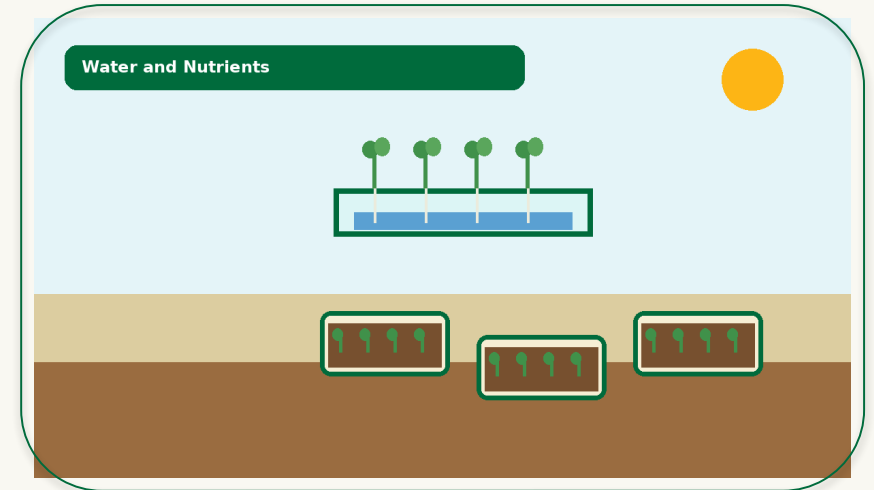
TEACHER PRACTICE

- Prepare a root-observation activity.

Water and Nutrients

Teacher knowledge and practical application

- Explain that soilless plants receive dissolved nutrients through managed water or nutrient solution.
- Only adults/authorized staff prepare concentrated nutrient solutions.
- Introduce concentration conceptually without requiring pupils to handle chemicals.
- Use clean water appropriate to the system.
- Record solution changes and crop observations.



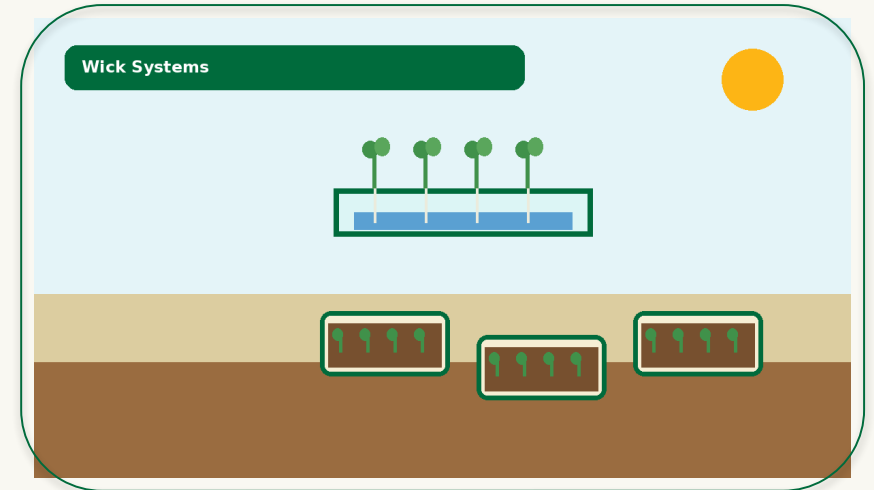
TEACHER PRACTICE

- Teacher task: explain the nutrient-water pathway safely.

Wick Systems

Teacher knowledge and practical application

- Explain how a wick can move water from a reservoir toward the growing medium by capillary action.
- Identify reservoir, wick, container and root zone.
- Keep the reservoir covered where possible.
- Observe moisture movement without adding dyes unless the system protocol permits safe demonstration.
- Check wicks for contact and blockage.



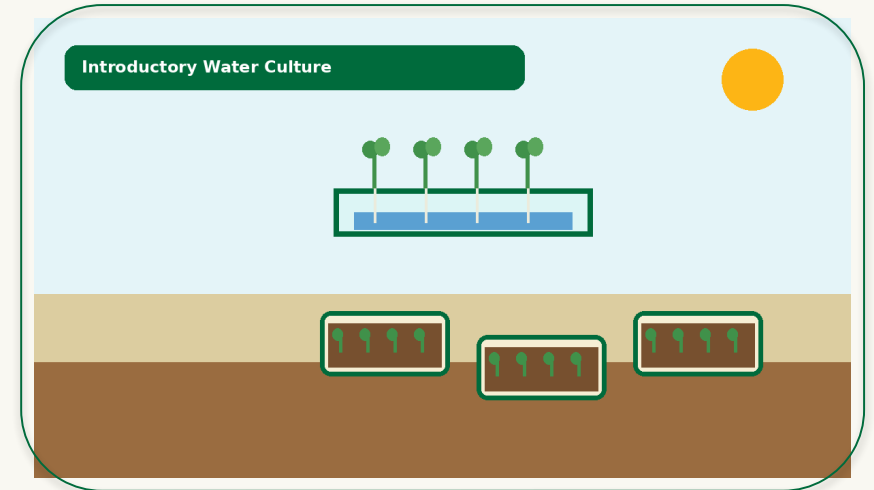
TEACHER PRACTICE

- Build or demonstrate a teacher-approved small wick unit.

Introductory Water Culture

Teacher knowledge and practical application

- Introduce water culture as roots growing in or above managed nutrient solution, depending on the system.
- Emphasize root oxygen needs and system hygiene.
- Use only stable, covered school units.
- Do not allow pupils to taste or handle nutrient solution.
- Observe roots through safe viewing points.



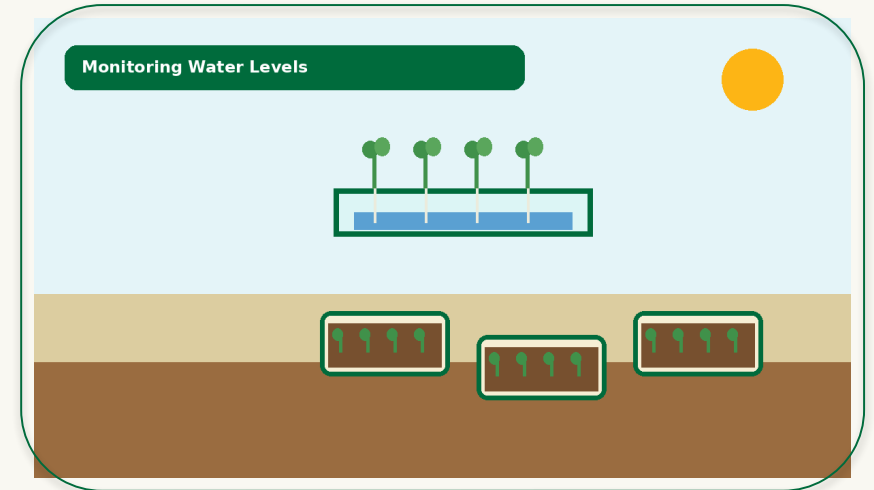
TEACHER PRACTICE

- Prepare a simple water-culture system diagram.

Monitoring Water Levels

Teacher knowledge and practical application

- Use marked reservoirs or level indicators to track water quantity.
- Record level at consistent times.
- Explain that falling level can result from plant uptake and evaporation, while leaks must also be checked.
- Refill according to the system procedure.
- Keep electrical equipment, if any, outside pupil handling.



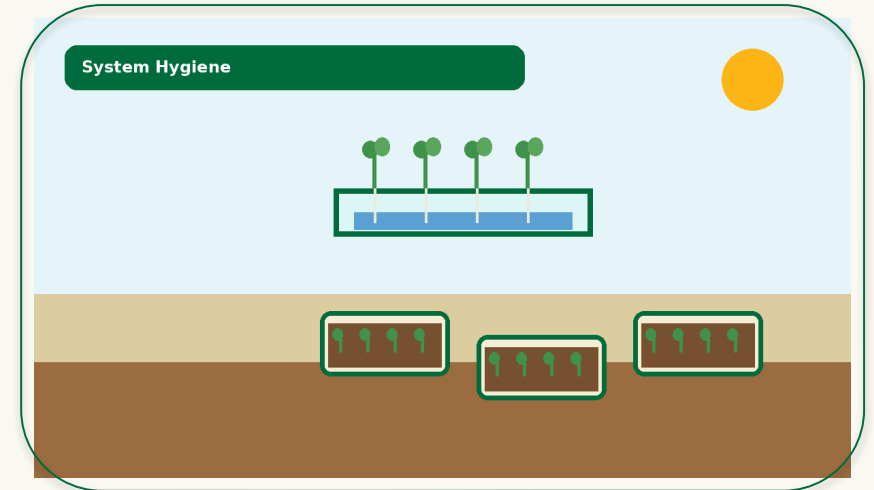
TEACHER PRACTICE

- Create a reservoir-level log.

System Hygiene

Teacher knowledge and practical application

- Keep reservoirs, channels, containers and tools clean according to the system schedule.
- Remove dead roots/leaves promptly.
- Prevent algae, mosquito access and contaminated inputs through good design and maintenance.
- Wash hands after handling plants and equipment.
- Adults manage nutrient concentrates, electrical devices and disinfection chemicals.



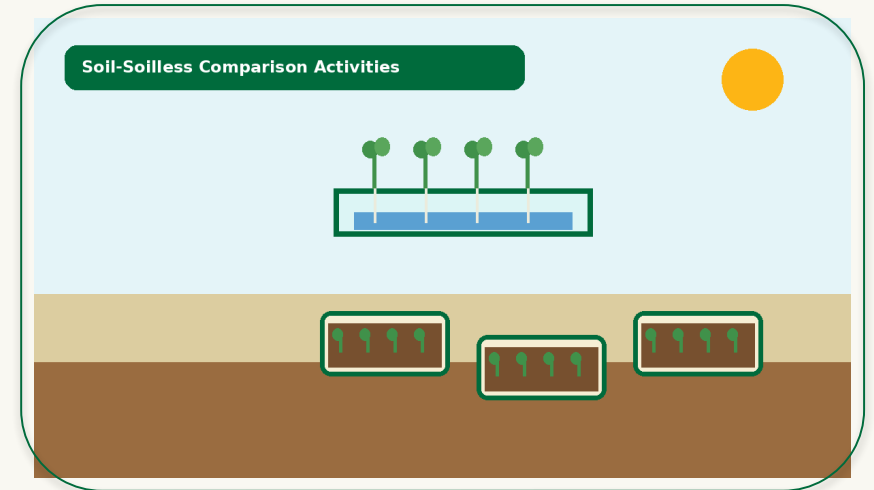
TEACHER PRACTICE

- Prepare a soilless hygiene checklist.

Soil-Soilless Comparison Activities

Teacher knowledge and practical application

- Grow the same crop in soil and an approved soilless unit when feasible.
- Define what will be compared: emergence, height, leaf number, water use or harvest.
- Keep measurements consistent and record management differences.
- Do not claim one system caused every observed difference without considering conditions.
- Use the activity to teach evidence-based comparison.



TEACHER PRACTICE

- Completion task: design a soil-versus-soilless class investigation.